



Dow Packaging & Speciality Plastics Product Data Sheet

BYNEL™ 22E757

Adhesive Resin

Description			
Product Description	BYNEL™ Series 2200 resins are modified ethylene acrylate resins. They contain a temperature stable ester which makes them functional in high temperature coextrusions. They are available in pellet form for use in conventional extrusion and coextrusion equipment designed to process polyethylene (PE) resins.		
Restrictions			
Material Status	Commercial: Active		
Typical Characteristics			
Characteristics / Benefits	Physical properties of BYNEL™ Series 2200 resins are typical of polyethylene/acrylate copolymer resins with similar density and melt index values.		
Applications	BYNEL™ 2200 series resins adhere to a wide variety of materials. They are most often used to adhere to PE, PP, PET and paper.		
Typical Properties			
Physical	Nominal Values		Test Method(s)
*Density ()	0.94 g/cm³	ASTM D792	ISO 1183
*Melt Flow Rate (190°C/2.16kg)	8.0 g/10 min	ASTM D1238	ISO 1133
Thermal	Nominal Values		Test Method(s)
*Melting Point(DSC)	92 °C (197.6 °F)	ASTM D3418	ISO 3146
Vicat Softening Point ()	54 °C (129.2 °F)	ASTM D1525	ISO 306
Adhesive Evaluation	The performance of any adhesive resin should be evaluated within the context of the application. The adhesive is designed to bond materials that would not ordinarily adhere to each other. In most cases, peel strength is used as a measure of performance. Although this is a convenient test, peel strength is affected not only by adhesion, but also by peel angle, separation rate, temperature, and tensile and modulus properties of the materials, and often by the time elapsed since the formation of the bond. Post-treatment of the multi-layer structure, such as heat sealing, thermoforming or orientation can also affect peel strength. If peel strength is used as a measure of adhesive performance, it is imperative that peel strength be evaluated not only at the time of manufacture, but throughout the life of the product and under all the various conditions to which the structure will be exposed. Only then can the performance of the adhesive be related to peel strength.		
Processing Information			
*Maximum Processing Temperature	310 °C (590 °F)		
General Processing Information	The temperature profiles shown below are for initial evaluations of BYNEL™ adhesive resins in the 2200 series. These profiles are designed to provide adequate exposure time of the adhesive resin to elevated temperatures. Because the BYNEL™ 2200 Series resins have low softening points, it is a good idea to run the rear of the extruder as cool as possible, then build quickly to the melt temperature. Water cooling of the screw and/or hopper feed throat may help avoid bridging problems. Specifically, in coextrusions with thermally sensitive resins such as EVOH or EVA, we suggest that the maximum melt temperature be limited to 235C (455F) to guard against overheating the EVOH or EVA. If adhesion results are adequate, we suggest evaluating even lower melt temperatures such as 210C - 220C (410F - 428F). For coextrusion with polyamides, polyolefins or other thermally stable resins, the melt temperature can be higher. We suggest a maximum melt temperature of 310C (590F). This should provide acceptable bond strengths and film quality under almost all coextrusion conditions. If adhesion results are adequate, melt		

temperatures can be lowered. While it is possible to extrude BYNEL™ 2200 series resins as high as 310C (590F), such high extrusion temperatures, particularly when coupled with long residence times, may result in some film imperfections. In certain streamlined extrusion operations, where residence times are short, it may be possible to use temperatures higher than 285C (545F).

Variation of these suggested temperature profiles may be appropriate depending upon the screw configuration, potential extruder horsepower limitations, potential back pressure limitations, the need to match rheologies and/or the stability of the other resins in the coextrusion. Film quality will also depend upon the residence time of the adhesive resin in the system. Dead spots may result in localized overheating and should be avoided by ensuring the flow path for the adhesive is as streamlined as possible.

We suggest using any standard polyolefin working screw when extruding BYNEL™ 2200 series resins. Excessively deep flights should be avoided as they might result in poor melting of the adhesive resin. It is also important to properly size the extruder for the output desired. Running large extruders at very low RPMs should be avoided.

For producing monolayer adhesive films with BYNEL™ 2200 adhesive resins, extrusion conditions commonly used for converting ethylene acrylate resins into films can be employed.

When extruding BYNEL™ 2200 series resins as an exposed outer surface in a multi-layer coextrusion, problems related to the tackiness and high coefficient of friction of these products may be evident. In this case, it is suggested that the extrusion temperature be lowered to 160C - 210C (320F - 410F) or less. Addition of slip and silica-based antiblock packages may also be appropriate to prevent blocking and improve film handling, although these additive packages may modify the resin's bonding characteristics.

If the coextrusion process is stopped for short periods of time, the screw in the adhesive extruder should be kept turning at a low RPM level. For a permanent shutdown, the BYNEL™ adhesive resin should be purged out using an available polyethylene resin run at the same extrusion temperature used during the extrusion process of the adhesive resin. Making frequent changes in screw speed during the shutdown process and subsequent start-up will help remove the previous material from the system more effectively. Sometimes upon start-up of the adhesive resin, excessive amounts of gel may be observed. This may be due to the natural ability of the adhesive resin to act as a purging compound. In this case, continued extrusion will eventually clear up the problem.

Extrusion Coating/Lamination Processing

Nominal Values

Processing Information

Proposed Extruder Set Temperatures

Feed Zone	160 °C (320 °F)
Second Zone	210 °C (410 °F)
Third Zone	260 °C (500 °F)
Fourth Zone	285 °C (545 °F)
Fifth Zone	285 °C (545 °F)
Adapter Zone	285 °C (545 °F)
Die Zone	285 °C (545 °F)

FDA Status Information

BYNEL™ 22E757 Adhesive Resin complies with Food and Drug Administration Regulation 21 CFR 177.1340(a) - - Ethylene-methyl acrylate copolymer resins, subject to the limitations and requirements therein. This Regulation describes polymers that may be used in contact with food, subject to the finished food-contact article meeting the extractive limitations under the intended conditions of use, as shown in paragraph (b) of the Regulation.

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Safety & Handling

For information on appropriate Handling & Storage of this polymeric resin, please refer to the material Safety Data Sheet.

A Product Safety Bulletin, material Safety Data Sheet, and/or more detailed information on extrusion processing and/or compounding of this polymeric resin for specific applications are available from your Dow representative.

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Published August 2019

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Version: 142.0

Last modified at 8/11/2019 3:37 PM